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3.

(1) $w'' - w' + \frac{1}{w} w = 0$

$\Rightarrow z=0$ 为方程奇点

令 $w = (z-z_0)^\rho \sum_{k=0}^{\infty} c_k (z-z_0)^k$

代入,

$$\begin{cases} \rho(\rho-1)c_0 = 0 \\ (k+\rho+1)(k+\rho)c_{k+1} - (k+\rho-1)c_k = 0 \end{cases}$$

$\Rightarrow \rho_1 = 0, \rho_2 = 1$

令 $\rho=1$ 代入,

$$c_k = \frac{k-1}{(k+1)k} c_{k-1}$$

$\Rightarrow$ $c_1 = c_2 = c_3 = \cdots = 0$

$\therefore$ $w_1 = c_0 z$

又 $w_2 = w_1(z) \int \frac{A}{w_1(z)^2} e^{-\int P(z)dz} dz$

$= c_0 z \int_0^z \frac{A}{c_0^2 z^2} e^z dz$

$= \frac{1}{c_0} z \int \frac{A}{z^2} e^z dz$

$= \frac{1}{c_0} z \int \frac{A}{z^2} \sum_{k=0}^{\infty} \frac{z^k}{k!} dz$

$= d_0 \left[\sum_{k=2}^{\infty} \frac{z^k}{k!(k-1)} + z \ln z - 1 \right]$

(1)、令 $w = z^{\rho}\sum_{k=0}^{\infty} c_k z^k$

代入，

$$\begin{cases} c_0\rho(\rho-1)=0 \\ c_k(k+\rho)(k+\rho-1)+c_{k-1}(k+\rho-1)=0 \end{cases}$$

即：$\rho=0$ 或 $\rho=1$

将 $\rho=1$ 代入，得：

$$c_k = \frac{-1}{k}c_{k-1}$$

$$\Rightarrow \quad c_n = \frac{(-1)^k}{k!}c_0$$

即：

$$w_1 = z\sum_{k=0}^{\infty}\frac{(-1)^k}{k!}c_0 z^k = c_0 z e^{-z}$$

则 w_2 用 w_1 表示，

$$w_2 = w_1\int\frac{A}{w_1^2}e^{-\int P(x)dx}dx$$

$$= c_0 z e^{-z}\int\frac{e^z}{z^2}dz$$

$$= c_0\left[\sum_{k=3}^{\infty}\frac{z^k}{k!(k-2)} - \frac{1}{2} - z + \frac{z}{2}\ln z\right]e^{-z}$$

$$= c_0\left[\sum_{j=0}^{\infty}\frac{(-1)^j}{j!}z^j\sum_{i=3}^{\infty}\frac{z^i}{i!(i-2)} + \left(\frac{z}{2}\ln z - z - \frac{1}{2}\right)\sum_{i=0}^{\infty}\frac{(-1)^i}{i!}z^i\right]$$